

Empirical Research Methods

— Assignment 3 —

University of St. Gallen

Foreword In this problem set, we will continue our analysis from Exercise 3 and further investigate the data in `distances_birthtown.csv`. In our data set, we observe individuals who are residents of *Country A* which borders *Country B*. *Country A* is a middle-income country while *Country B* is high-income country with an especially strong medical industry. You are hired by a university in Country A that would like to better understand what factors influence an individual's chosen career path. Specifically, they believe that being from a high-income family increases the probability that a student pursues a career in the medical field.

Make sure to download the updated data: `distances_birthtown_up.csv`

For each question, and if not otherwise specified, keep your answer under 150 words (0.5 points per question are removed if the answer is longer). A maximum of 5 points can be reached.

For clarifications, please contact: federica.mascolo@unisg.ch

Table 1: Description of Variables

Variable name	Description	Comments
<i>id</i>	Individual-ID	
<i>region</i>	Code for region of town of birth	Values: 1-15
<i>travel_time</i>	Travel time in hours to Country B from town of birth	
<i>income</i>	Annual income of individual in \$1,000	Unreported income: -9999999
<i>income_p1</i>	Income of parent 1 in \$1,000	Unreported income: -9999999
<i>income_p2</i>	Income of parent 2 in \$1,000	Unreported income: -9999999
<i>medicine</i>	Indicator for working in medical industry	Values: “Yes”, “No”

1 Exercise 1

In this question, you will derive how OLS coefficients change under simple transformations of the regressors. You can typeset your answers directly using the “math-mode” functionality in R Markdown. See [this math-mode guide](#) for a cheatsheet on how to write equations in R Markdown.

Suppose that you observe the variables $Y_{i,1}$, $Y_{i,2}$, and X_i , and estimate the following simple linear regressions:

$$Y_{i,1} = \beta_0 + \beta_1 X_i + u_i$$

$$Y_{i,2} = \alpha_0 + \alpha_1 X_i + v_i$$

Now define the following transformed variable: $Q_i = X_i - 10$. Derive the requested relationship between the OLS coefficients.

1. Let $\hat{\delta}_1$ be the slope coefficient obtained from estimating

$$Y_{i,1} = \delta_0 + \delta_1 Q_i + u_i.$$

Show that

$$\hat{\delta}_1 = \hat{\beta}_1.$$

What happens to the OLS coefficients when a constant is added to or subtracted from a regressor? Does the relationship between $\hat{\delta}_0$ and $\hat{\beta}_0$ changes? [**1 point**]

2 Exercise 2

1. Implement a regression analysis to jointly investigate the relationship of *medicine* with *travel_time* and *income_parents* :
 - (i) Import the dataset, explore it, and clean it if necessary. Generate the variable:
 $income_parents = income_p1 + income_p2$.

- (ii) Write a regression equation to jointly investigate the relationship of *medicine* (as the outcome variable) with *travel_time* and *income_parents* (as the independent variables). Write the interpretation for each of the coefficient parameters. You may assume that MLR.1-MLR.5 hold. **[1 point (200 words or less)]**

- (iii) Now estimate your regression you wrote in Exercise 2(ii) and the following regression:

$$medicine_i = \delta_0 + \delta_1 income_parents_i + v_i. \quad (1)$$

Compare the point estimates from your specification in part (ii) to those from (iii). Are they statistically and economically different? And if so, briefly provide a possible explanation as to why.

[1 point (200 words or less)]

- (iv) Do you think using the variable *region* instead of *travel_time* would help to explain better our research question? Explain your answer. **[0.5 point (200 words or less)]**
- (v) Describe and implement a test for whether *income_parents* is correlated with *travel_time*. Clearly state the null and alternative hypotheses and how you would construct the test statistic. What is the result of your test? What do you conclude from this in terms of your specified hypotheses? **[1 point (200 words or less)]**
- (vi) Consider the regression specification of Exercise (ii). Test whether *travel_time* and *income_parents* are jointly significant. Clearly state the null and alternative hypotheses and explain how you would construct the test and the results. **[1 point]**

Tables

Figure 1: Critical values of the F distribution with $\alpha = 5\%$

5% Critical Values of the F Distribution											
Numerator Degrees of Freedom											
		1	2	3	4	5	6	7	8	9	10
D e n o m i n a t o r D e g r e e s o f F r e e d o m	10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98
	11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85
	12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75
	13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67
	14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60
	15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54
	16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49
	17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45
	18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41
	19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38
	20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35
	21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32
	22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30
	23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32	2.27
	24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25
	25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28	2.24
	26	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27	2.22
	27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25	2.20
	28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24	2.19
	29	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22	2.18
	30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16
	40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08
	60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	1.99
	90	3.95	3.10	2.71	2.47	2.32	2.20	2.11	2.04	1.99	1.94
	120	3.92	3.07	2.68	2.45	2.29	2.17	2.09	2.02	1.96	1.91
	∞	3.84	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88	1.83

Figure 2: Critical values of the t distribution

Critical Values of the t Distribution						
Significance Level						
1-Tailed: 2-Tailed:		.10 .20	.05 .10	.025 .05	.01 .02	.005 .01
D e g r e e s o f F r e e d o m	1	3.078	6.314	12.706	31.821	63.657
	2	1.886	2.920	4.303	6.965	9.925
	3	1.638	2.353	3.182	4.541	5.841
	4	1.533	2.132	2.776	3.747	4.604
	5	1.476	2.015	2.571	3.365	4.032
	6	1.440	1.943	2.447	3.143	3.707
	7	1.415	1.895	2.365	2.998	3.499
	8	1.397	1.860	2.306	2.896	3.355
	9	1.383	1.833	2.262	2.821	3.250
	10	1.372	1.812	2.228	2.764	3.169
	11	1.363	1.796	2.201	2.718	3.106
	12	1.356	1.782	2.179	2.681	3.055
	13	1.350	1.771	2.160	2.650	3.012
	14	1.345	1.761	2.145	2.624	2.977
	15	1.341	1.753	2.131	2.602	2.947
	16	1.337	1.746	2.120	2.583	2.921
	17	1.333	1.740	2.110	2.567	2.898
	18	1.330	1.734	2.101	2.552	2.878
	19	1.328	1.729	2.093	2.539	2.861
	20	1.325	1.725	2.086	2.528	2.845
	21	1.323	1.721	2.080	2.518	2.831
	22	1.321	1.717	2.074	2.508	2.819
	23	1.319	1.714	2.069	2.500	2.807
	24	1.318	1.711	2.064	2.492	2.797
	25	1.316	1.708	2.060	2.485	2.787
	26	1.315	1.706	2.056	2.479	2.779
	27	1.314	1.703	2.052	2.473	2.771
	28	1.313	1.701	2.048	2.467	2.763
	29	1.311	1.699	2.045	2.462	2.756
	30	1.310	1.697	2.042	2.457	2.750
	40	1.303	1.684	2.021	2.423	2.704
	60	1.296	1.671	2.000	2.390	2.660
	90	1.291	1.662	1.987	2.368	2.632
	120	1.289	1.658	1.980	2.358	2.617
	∞	1.282	1.645	1.960	2.326	2.576